Budget-Aware Self-Configuring Neural Architecture Search for Unseen Data

Sebastian Seifert, Luis Melzer & Leander von Gottberg

1 Motivation

The NAS Unseen-Data Challenge (AutoML 2026) evaluates a fully automated pipeline on *hidden* classification datasets under a strict wall-clock budget per dataset. Inputs arrive as image-shaped tensors but range from natural and satellite images to *encoded* data (rendered text, Sudoku boards, chess positions) where the spatial prior of a CNN misleads. All decisions must be made at runtime, blind.

Our starting point is the winning method of the 2025 edition, EGNAS [1], a macro–micro search (depth, width, per-layer operation) over convolutional blocks. Reproducing it surfaced two weaknesses. First, rankings from short training runs are unreliable: operations that lead early often lose after full training, so fine-grained (micro) operation search tends to *harm* final accuracy. Second, the space is confined to one family: in our experiments a *fixed* 2-D dilated-convolution backbone (temporal convolutional networks [3]) beat the fully searched CNN on most datasets, while plain CNNs and random architectures won on the encoded ones, matching random search as a strong NAS baseline [2, 4]. The leverage is choosing the right architecture *family* per dataset, not operation search.

2 Research Questions

RQ1. Can macro–micro NAS [1] be generalised across heterogeneous architecture families (plain CNNs, 2-D dilated and 1-D temporal convolution networks), making families *comparable* within a single time-budgeted search stage despite the known unreliability of low-fidelity rankings [5]? **RQ2.** Can cheap, training-free measurements of the raw data decide at runtime which architectural prior fits an unseen dataset? **RQ3.** How can harmful data augmentation be detected at runtime, given that augmentation acts as regularisation (it costs accuracy early and pays late) and that, as we observe, its decisions do not transfer between architecture families?

3 Approach

We propose a self-configuring pipeline that measures the data first, gates every decision, and spends all remaining time on training (Fig. 1, p. 2). At its core is one search principle applied at every level: each choice is searched against a robust incumbent default, scored by a short-training proxy, and adopted only when it beats the default beyond the proxy noise *and* survives a longer verification re-training. The search is reliability-aware by design, never trusting a short-training ranking alone, the failure mode that makes naive operation search harmful. (1) *Measure*: a fingerprint computed directly from the raw input tensors, with no training involved. The mean correlation of neighbouring pixels separates the two data families with a wide margin: between -0.05 and 0.07 on the four encoded public datasets versus 0.38 to 1.00 on all others. (2) *Family selection*: image-like data goes to the dilated backbone directly; encoded data, where the prior is ambiguous, evaluates all families at matched wall-clock and adopts the leader only when it clears the default beyond the ranking noise, otherwise *abstaining* to the default (a built-in null hypothesis that bounds the downside). (3) *Augmentation probe*: a two-arm test run *after* the family choice, with arms of the chosen family; it deviates only on a clear margin (RQ3). (4) *Structural search*: depth and width are searched against an incumbent default, adopting a non-default choice only when it wins beyond the proxy noise *and* survives re-training verification at a longer horizon. (5) *Operation-level exploration*: on encoded data with budget to spare, the search samples architectures from a broad operation vocabulary under the same verify-before-adopt rule; random sampling is a deliberate, competitive NAS baseline [4]. (6) *Train to budget*: the winner trains on the full remaining time with a cosine learning-rate schedule [6] sized to reach zero exactly when the budget ends. All stage budgets are fractions of the clock, so the pipeline degrades gracefully from many hours down to minutes.

4 Required Steps

(1) Reproduce the EGNAS baseline and probe candidate architecture families at full budget (done). (2) Validate each decision mechanism in isolation against known full-budget outcomes. (3) Integrate the pipeline and verify all execution paths, including low-budget collapse. (4) Full multi-seed evaluation. (5) A budget-ladder study (0.5h to 8h) measuring which stages pay at which budget. (6) A generalisation test on untouched external datasets. (7) Packaging and stress tests.

5 Evaluation Strategy

Primary evaluation uses the 13 public competition datasets with the official adjusted score (accuracy versus a per-dataset benchmark, capped), at least three seeds for final configurations. Baselines: the reproduced EGNAS search, each fixed family, and a best-family oracle bounding the achievable routing gain. Every stage writes its measurements, margins, and choices to a per-dataset decision record, so gains and losses are attributable to individual decisions. Robustness is evaluated across budgets (30 minutes to 12 hours), on the external suite, and against published competition bests [2]. First results of the current pipeline, including the configuration it selects per dataset, are shown in Fig. 2 (p. 3).

6 Potential Extensions

The pipeline scales with the budget: at large budgets it admits extra families (recurrent and attention blocks) via the gated comparison and searches per-stage *heterogeneous* architectures rather than one uniform stack; on ties it trains a cross-family ensemble, kept only if it beats the best single model. We also build competition-style datasets blind to the pipeline as a generalisation test.

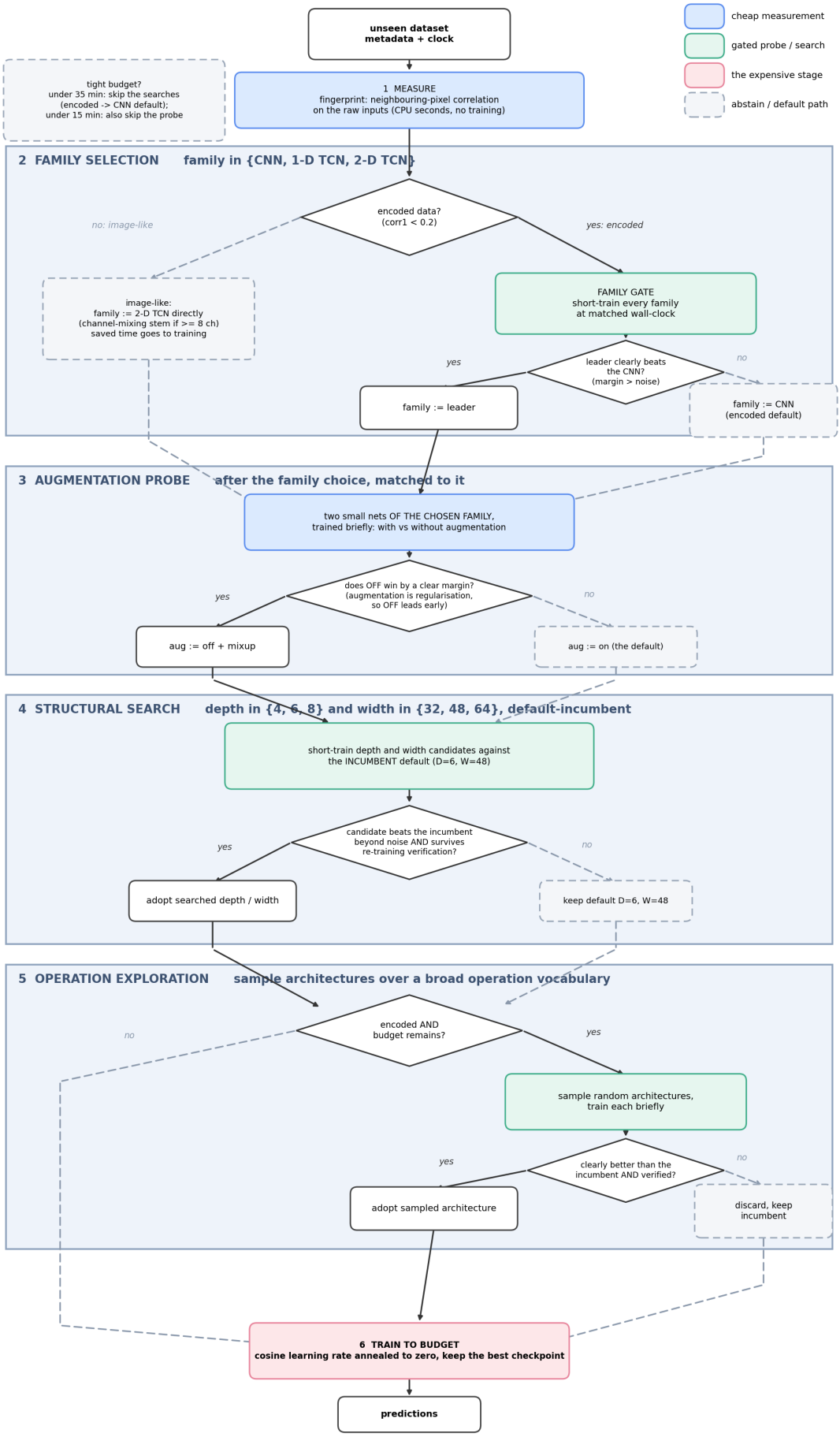


Figure 1: The decision pipeline, run once per unseen dataset. Diamonds are runtime decisions, dashed paths are the abstain-to-default routes.

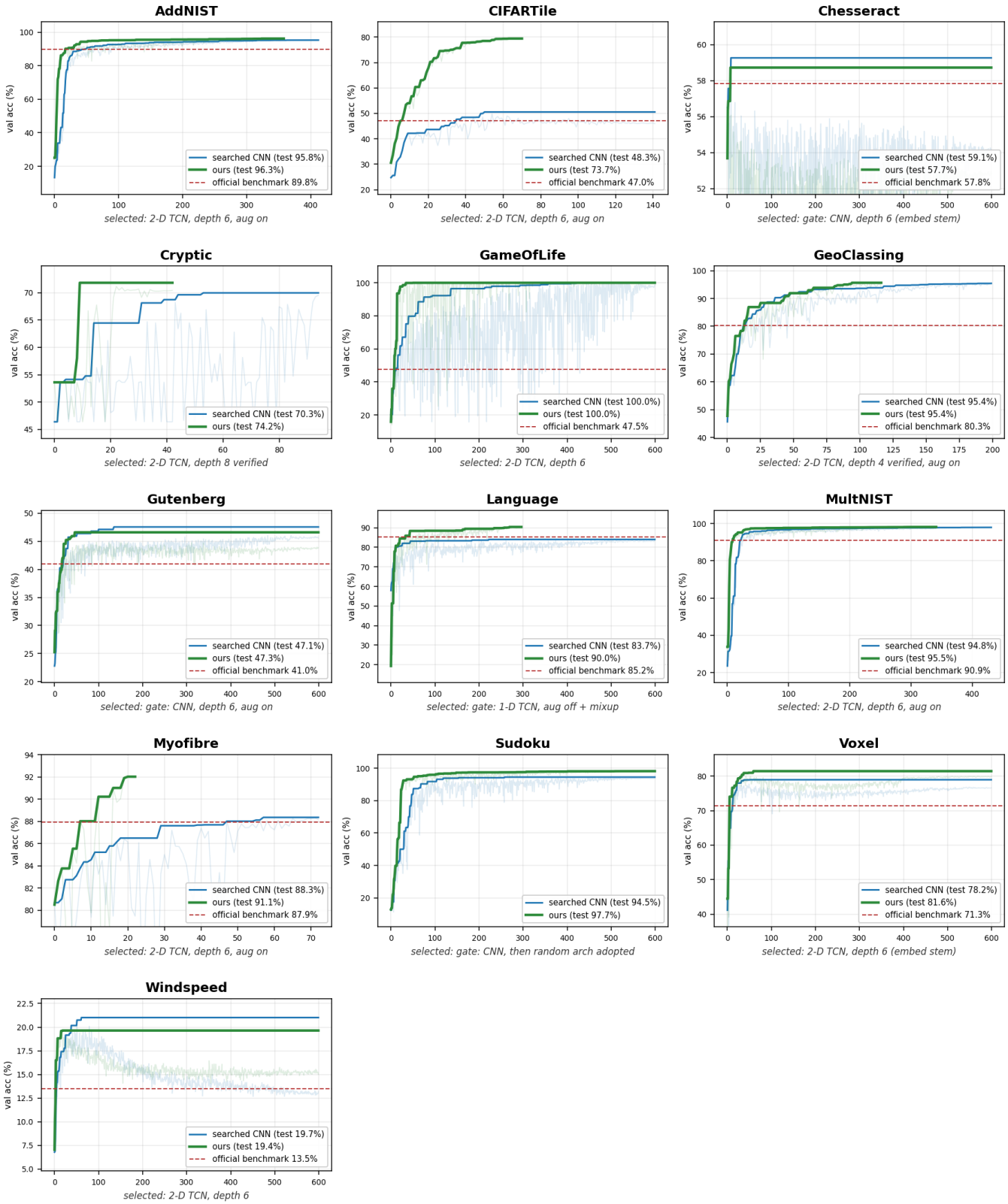


Figure 2: Our current best approach versus the searched CNN (our reproduction of the winning macro–micro NAS [1] with identical training improvements), 13 public datasets, single seed, 2 h per dataset on a single GB10. Curves: validation accuracy (faint per-epoch, solid best-so-far). Legends: final test accuracy, which can differ from validation. Red dashed: the official benchmark from the dataset metadata; unpublished for Sudoku and Cryptic. The selected configuration is noted under each panel. Totals (adjusted score): ours **+62.5**, searched CNN +48.2.

References

[1] S. Siddiqui, C. Kyrkou, T. Theocharides. *Efficient Global Neural Architecture Search*. SN Computer Science 6:282, 2025.

[2] R. Geada, D. Towers, M. Forshaw, A. Atapour-Abarghouei, A. S. McGough. *Insights from the Use of Previously Unseen Neural Architecture Search Datasets*. CVPR 2024.

[3] S. Bai, J. Z. Kolter, V. Koltun. *An Empirical Evaluation of Generic Convolutional and Recurrent Networks for Sequence Modeling*. arXiv:1803.01271, 2018.

[4] L. Li, A. Talwalkar. *Random Search and Reproducibility for Neural Architecture Search*. UAI 2019.

[5] C. White, A. Zela, B. Ru, Y. Liu, F. Hutter. *How Powerful are Performance Predictors in Neural Architecture Search?* NeurIPS 2021.

[6] I. Loshchilov, F. Hutter. *SGDR: Stochastic Gradient Descent with Warm Restarts*. ICLR 2017.